ResearchOS: Illustrated Evidence-Grounded Multi-Agent Architecture

A human-guided research workspace where specialized agents discover, reason, evaluate, verify, and write from traceable evidence

Researcher + Project Workspace
goal, documents, data, constraints, feedback

request + context

External Research Services
models, scholarly search, tools, code runtime

interact

RESEARCHOS AGENT RUNTIME

Coordinated research with evidence and verification at every stage

1 Evidence Workspace

Documents + Papers

Project Library

Document Intelligence
parse - chunk - embed - bind provenance

Hybrid Retrieval
dense + lexical + ranked evidence packs

evidence

Literature + Gap Agents
search, deduplicate, synthesize
map claims and unresolved gaps
DISCOVER

Hypothesis + Lineage Agents
generate falsifiable candidates
track ancestry and rank proposals
HYPOTHESIZE

Novelty + Feasibility Agents
compare prior art and constraints
check data, compute, risk, ethics
EVALUATE

Critic + Writing Agents
challenge assumptions and revise
compose evidence-linked sections
SYNTHESIZE

ResearchOS Coordinator
compile - plan - route - monitor - replan

Experiment + Statistics Agent
execute code and capture provenance
review methods, effects, uncertainty
TEST

Verify + Deliver

Claim and citation validation

Draft + references + export

IEEE - APA 7 - Harvard - MLA 9

verification failures return claims for retrieval, revision, or replanning

PERSISTENT RESEARCH MEMORY + GOVERNANCE

PostgreSQL State
projects, runs, conversations

Evidence + Vector Index
chunks, embeddings, provenance

Artifact Store
sources, drafts, results, exports

Security + Audit
tenant scope, approvals, usage

Solid arrows: workflow and delivery - Teal arrows: evidence flow - Coral dashed arrow: quality-control feedback - Gray dashed lines: routing and persistence